Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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1. Team & Project Overview

Team Name	Here We Go
Selected Domain	Domain 1
Selected SDG Track (1–17)	Main 4, Sub: 5, 8
Project / Solution Name	Ikiguy.
Name (Leader)	Abel Nathanael Hutapea
Email & Phone (Leader)	DSC2309294@xmu.edu.my & 60-14760-1017
University Affiliation(s)	Xiamen University Malaysia
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

In recent times, artificial intelligence (AI) has largely improved, to the point where not only it is a useful tool, but also a threat to replace human beings in jobs, especially those who are new to the job market, such as fresh graduates and juniors. This is due to the ability of AI to carry out easier tasks not only at a faster pace but also at a larger scale. While the potential downsizing that can arise from AI may benefit companies, in the long run, it can cause severe harm to the society and economy.

While the present effects of AI displacing jobs may be relatively negligible, it is the unpredictable advancement of AI that is worrying. For example, LLMs, which were first introduced in 2018 with GPT-1, frequently hallucinated, but in the year of 2026, LLMs, through agentic AIs, can perform complex tasks such as helping with research or book hotels. This shows that AI has the potential to quickly evolve, overtaking the capabilities of humans in some domains.

As humans have learned to live with technologies such as electric, and the internet, humans must learn to live with the existence of AI. One approach to this is for humans to improve themselves to not get displaced by AIs. Therefore, we aim to provide a solution to this issue by helping to those who are looking to find jobs by guiding them on how they can improve themselves to be more attractive to employers.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

FutureReady AI is an AI-assisted career navigation application that helps students, graduates and early-career professionals understand their skills, explore suitable careers and prepare for changes driven by artificial intelligence. Users upload a synthetic CV or enter a profile, review the extracted skills, and receive explainable career matches, identified skill gaps and an affordable learning roadmap.

The application uses generative AI to estimate skill proficiency from CV evidence and interpret dated research about how AI may change workplace tasks. These interpretations support career outlooks and clearly labelled hypothetical scenarios, helping users explore different futures without treating predictions as facts. Career matching uses transparent scoring against standardized occupational requirements, while learning roadmaps connect missing skills to verified resources and relevant UN Sustainable Development Goals. Users can review and correct AI-generated assessments before continuing.

We intend to apply skills covered in the SDC Phase 1 workshops, including GenAI API integration, structured data architecture, backend API development and cloud hosting. The prototype combines Groq's GPT-OSS-120B with RunPod-hosted GLM-4.7-Flash and Gemini as fallback models, a FastAPI backend, structured JSON catalogs and a React interface. Replit is the planned deployment platform. Together, these tools support a practical application that turns career uncertainty into understandable options and actionable learning steps.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

FutureReady AI will use React, TypeScript, Vite and Tailwind CSS for its interactive frontend, supported by a Python backend using FastAPI, Pydantic and Uvicorn. CV processing uses pypdf, python-docx and RapidOCR/ONNX Runtime to extract text from PDFs, Word documents and scanned images. Groq's GPT-OSS-120B provides AI-assisted skill-level estimation and research interpretation, with RunPod-hosted GLM-4.7-Flash and Google Gemini as fallback models.

The prototype stores occupations, skills, research evidence, courses and SDG mappings in versioned JSON catalogs, while temporary user profiles use expiring in-memory storage. Its data sources include O*NET 31.0, Google Research and Microsoft Research feeds, verified course resources and UN SDG references. Deterministic algorithms provide explainable career matching, skill-gap analysis and learning priorities.

The architecture follows this flow: synthetic CV or profile → React interface → FastAPI backend → CV parsing and user skill review → O*NET-based role matching → evidence-informed AI outlook and scenarios → personalized learning roadmap with SDG connections. The backend calls the AI providers when interpretation is needed. Replit is the planned host for both the frontend and backend, while GitHub Actions supports scheduled evidence discovery.

The project utilized the sponsor-provided RunPod cloud platform. We deployed GLM-4.7-Flash on a serverless A100 GPU endpoint and verified inference using a fictional CV. RunPod provides backup inference for skill assessment and research processing. Public application deployment on Replit remains pending.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

Our team applied design thinking by focusing on a practical user question: “Which careers fit my current skills, and what should I learn next?” We defined the problem around unclear career recommendations, changing skill requirements and limited access to affordable training. During ideation, we prioritized explainable career matches, editable skill profiles and actionable learning roadmaps.

We translated these priorities into a working local prototype with a guided journey: upload a synthetic CV, review extracted skills, explore suitable roles and examine learning options. Users retain control over AI-estimated skill levels, while dated evidence and clearly labelled hypothetical scenarios communicate uncertainty. Testing with synthetic profiles checks document extraction, recommendation consistency and the complete user flow without requiring personal CVs.

Technical viability comes from a modular React and FastAPI architecture, structured data contracts and fallback AI providers. Deterministic matching and cached learning resources keep core functions available when AI services fail.

For Phase 2, we will deploy the application on Replit, validate the public user journey, expand verified evidence and course coverage, and conduct usability sessions with target users. Their feedback will guide improvements to navigation, explanations and accessibility before broader deployment.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Target 4.4 "By 2030, substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs and entrepreneurship." Target 5.b "Enhance the use of enabling technology, in particular information and communications technology, to promote the empowerment of women." Target 8.6 "By 2020, substantially reduce the proportion of youth not in employment, education or training."
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	The current unemployment rate of Malaysia is 2.9% (as of August 2026), while the current unemployment rate of youth (15 to 24 years old) in Malaysia is 10.1%.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	% increase in employment rate as a whole and of the youth

Metric Category	Your Response
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	0.5% decrease in unemployment in Malaysia, and a 5% decrease in unemployment of youth in Malaysia

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React and React DOM	Code - frontend framework	MIT
TypeScript	Code - programming language and compiler	Apache 2.0
Vite	Code - frontend build tool	MIT
Tailwind CSS	Code - styling framework	MIT
FastAPI	Code - backend framework	MIT
Pydantic	Code - data validation	MIT
Uvicorn	Code - application server	BSD 3-Clause
pypdf	Code - PDF processing	BSD 3-Clause
python-docx	Code - Word document processing	MIT
Pillow	Code - image processing	MIT-CMU
pypdfium2 / PDFium	Code - PDF rendering	Apache 2.0 / BSD 3-Clause

		for pypdfium2; BSD-style and bundled third-party licenses for PDFium
RapidOCR	Code - optical character recognition	Apache 2.0
ONNX Runtime	Code - OCR model inference	MIT
vLLM	Code - model serving on RunPod	Apache 2.0
HTTPX	Code - HTTP client for testing	BSD 3-Clause
Playwright	Code - browser testing	Apache 2.0
Inter	Font asset - interface typography	SIL Open Font License 1.1
O*NET 31.0 Database	Dataset - occupations, skills, tasks and software associations	Creative Commons Attribution 4.0 International (CC BY 4.0)
OpenAI GPT-OSS-120B	Pre-trained model - accessed through Groq	Apache 2.0; Groq service terms also apply
Z.ai GLM-4.7-Flash	Pre-trained model - hosted on RunPod	MIT; RunPod service terms also apply
Google Gemini	Proprietary model - fallback API	Google API Terms of Service and Gemini API Additional Terms

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



24/10//2026